Grauert's ruled surfaces and semipositive metrics: Lipschitz existence and a C^1 obstruction

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Abstract

Let B be a smooth projective complex curve of genus at least two, let F be a positive-degree line bundle on B, and let $0 \neq \xi \in H^1(B, F)$. The affine F-torsor defined by ξ has a projective completion $q : X = \mathbb{P}_{\text{lines}}(E) \to B$, where $0 \to F \to E \to \mathcal{O}_B \to 0$ represents ξ. Its infinity section $C = \mathbb{P}_{\text{lines}}(F)$ has negative normal bundle. We give a detailed account of this construction and of the distinguished line bundle $L_0 = \mathcal{O}_X(C) \otimes q^*F$. We prove that L_0 is nef and big, has null locus C, and is not semiample. We then prove two regularity statements: L_0 admits a Hermitian metric with locally Lipschitz weights and positive curvature current, smooth with strictly positive curvature away from C; but it admits no such semipositive metric of class C^1, even near C. The existence proof uses a quantitatively controlled cusp in the norm of a defining section of C and a regularized maximum. The obstruction uses holomorphic graph tests for first derivatives of plurisubharmonic weights. The resulting bounded potentials have minimal singularities and answer, for both the original and modified examples, the bounded-potential question recorded in Zhang's December 2025 preprint. All regularity arguments are proved here. The account distinguishes them from Grauert's contraction construction, Koike's smooth first-obstruction theorem, and Zhang's calculation for the original example; it makes no claim about publication priority of the arguments presented here.

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1 Introduction and principal results

Grauert's construction in [Gra62, §4,8(d), pp. 365–366] exhibits a smooth projective ruled surface containing a negative section whose analytic contraction is not projective. The construction is governed by an additive cohomology class, not just by the numerical data of the section. This distinction is visible in the positivity of a particular line bundle on the ruled surface. Filip and Tosatti singled out this example in [FT18, Problem 2.2, p. 2989]. Koike's first-obstruction theorem and his modified example appear in [Koi21, Theorem 1.4 and Example 4.1]. Zhang treats the original degree-one construction in [Zha25, Theorem 3.2] and asks in [Zha25, §4] whether its distinguished class, or that of the modified example, has a positive current with bounded potentials and minimal singularities.

This manuscript works with the following data throughout its geometric applications. Let B be a smooth connected projective complex curve of genus $g \geq 2$, let F be a line bundle of degree $d > 0$, and choose $0 \neq \xi \in H^1(B, F)$. Let

$$0 \longrightarrow F \longrightarrow E \longrightarrow \mathcal{O}_B \longrightarrow 0 \tag{1.1}$$

be the extension represented by ξ, and put

$$X = \mathbb{P}_{\text{lines}}(E), \qquad C = \mathbb{P}_{\text{lines}}(F), \qquad L_0 = \mathcal{O}_X(C) \otimes q^*F. \tag{1.2}$$

Here projectivization parametrizes lines in each vector space E_b. The notation F, E is used for vector bundles and for their sheaves of holomorphic sections; the affine torsor itself is denoted by T. The construction and every transition convention are specified in Section 3.

Theorem 1.1. *For the data in (1.1)–(1.2) the following statements hold.*

(i) *The surface X is smooth and projective, $C \cong B$, and $N_{C/X} \cong F^{-1}$. In particular $C^2 = -d$.*

(ii) *The bundle L_0 is nef and big,*

$$L_0^2 = d, \qquad L_0 \cdot C = 0, \qquad L_0|_C \cong \mathcal{O}_C, \qquad \mathrm{Null}(L_0) = C.$$

Every section of every positive tensor power of L_0 vanishes along C; in particular L_0 is not semiample.

(iii) *There exists a positive definite Hermitian metric on L_0 with locally Lipschitz weights and semipositive curvature in the sense of currents. The metric can be chosen smooth with strictly positive curvature on $X \setminus C$. Its curvature has bounded local potentials and minimal singularities.*

(iv) *No neighborhood of C carries a C^1 Hermitian metric on L_0 whose curvature is a positive current. In particular there is no C^2 or smooth semipositive Hermitian metric on L_0.*

The proof is distributed as follows. The geometric statements and the absence of sections on the affine torsor are proved in Section 3. The obstruction in (iv) is Theorem 4.3. The metric in (iii) is constructed in Theorem 5.4, and its application to L_0 is given in Corollary 6.1. This organization makes the existence and nonexistence arguments independent: neither is deduced by a regularization or limiting argument from the other.

The contrast in Theorem 1.1 concerns the regularity of local metric weights. It is not a claim that every notion of differentiability between Lipschitz continuity and C^1 has been classified. In particular, existence of a Lipschitz metric does not supply a semipositive C^1 approximation in the same class.

Scope of self-containedness. All local analytic lemmas, the curvature inequalities, the first-jet obstruction, the metric gluing, and the minimal-singularity comparison are proved in detail. For the surrounding algebraic and analytic geometry we use standard foundational results: curve Riemann–Roch and Serre duality, solvability of the Poisson equation on a compact curve, the algebraic–analytic comparison for projective varieties, the Nakai–Moishezon criterion, the Hilbert polynomial of an ample line bundle, Grauert's contraction criterion, the Leray–Hirsch theorem, and the compact Kähler $\partial\bar\partial$-lemma. Each use is stated with its needed hypotheses and cited. These background theorems are not presented as new results or silently replaced by an appeal to a stronger assertion. In particular, the metric existence and obstruction proofs do not invoke a holomorphic tubular neighborhood theorem or a smooth semipositivity theorem.

2 Conventions and elementary analytic tools

2.1 Metric weights, curvature, and regularity

Let L be a holomorphic line bundle on a complex manifold S. A positive definite Hermitian metric h has, in a nowhere vanishing holomorphic frame e_i, a real weight φ_i defined by

$$|e_i|_h^2 = e^{-\varphi_i}.$$

If $e_i = g_{ij}e_j$, where g_{ij} is holomorphic and nowhere zero, then

$$\varphi_i = \varphi_j - \log|g_{ij}|^2. \tag{2.1}$$

The second term is smooth and pluriharmonic. Consequently continuity, local Lipschitz continuity, and membership in C^k of the weights are independent of the chosen holomorphic frames. For a continuous positive metric, its local positive norm functions and its weights have the same local Lipschitz or C^k regularity: on each relatively compact chart the norm is bounded above and bounded away from zero, and the logarithm and exponential have bounded derivatives on the corresponding ranges.

We fix the normalization

$$d^c = \frac{i}{4\pi}(\bar\partial - \partial), \qquad dd^c = \frac{i}{2\pi}\partial\bar\partial.$$

For a smooth metric, $c_1(L,h) = dd^c\varphi_i$ is a global real closed $(1,1)$-form representing $c_1(L)$. More generally, a locally integrable weight defines the curvature current by the same formula. The metric is *semipositive in the sense of currents* when its local weights are plurisubharmonic. For continuous weights, this is equivalent to positivity of their distributional complex Hessians. A singular metric may have weights with value $-\infty$; a continuous positive definite metric in this paper always has finite weights.

Fix a smooth reference metric h_0 and write $\theta = c_1(L,h_0)$. A global function u changes the squared metric by $h = h_0e^{-u}$ and changes the local weights by $\varphi_i = \varphi_{0,i} + u$. We write

$$\mathrm{PSH}(S,\theta) = \{u : \ \varphi_{0,i} + u \text{ is plurisubharmonic on every local frame chart}\}.$$

Equivalently $\theta + dd^c u \geq 0$ in currents, with the usual upper semicontinuous representative. This notation makes sense on an open subset as well as on a compact manifold.

For matrix computations, if a real $(1,1)$-form is written

$$\beta = \frac{i}{2\pi}\sum_{j,k}\beta_{j\bar{k}}dz_j \wedge d\bar{z}_k,$$

we denote its associated Hermitian form by $\beta_{V\bar{W}} = \sum_{j,k}\beta_{j\bar{k}}V^j\overline{W^k}$. This notation removes the factor $i/(2\pi)$ and does not mean the literal evaluation of the alternating two-form on $(V,\bar{W})$. Positivity of β is positivity of this Hermitian form. The frame used to evaluate it may be smooth rather than holomorphic.

2.2 Subharmonic limits, maxima, and removable sets

We record the elementary tools used below. They can also be found in the general treatments [Dem12b, Chapter I, §5] and [Dem12a].

Lemma 2.1. *A locally uniform limit of continuous subharmonic functions is subharmonic. If both a continuous real function and its negative are subharmonic, the function is harmonic. A continuous subharmonic function on a compact connected Riemann surface is constant.*

Proof. For a closed Euclidean disc contained in a coordinate domain, the submean inequality for each approximating function passes to the limit because convergence is uniform on the closed disc and its boundary. The limit is continuous and satisfies the submean inequality on every such disc, which is the characterization of subharmonicity for continuous functions. If the same inequality holds for a function and its negative, both inequalities are equalities. The function has the mean-value property and is harmonic. For completeness, convolving with a radial smooth approximate identity on a smaller disc leaves a function with the mean-value property unchanged: the convolution is a weighted average of its circular means. The convolution is smooth, so the function is smooth and its Laplacian is zero. Finally, a continuous function on a compact surface attains its maximum. The maximum principle for the submean inequality makes a subharmonic function constant on a neighborhood of a maximum point; the same argument and connectedness make it constant on the whole surface. □

Lemma 2.2. *Let D be a smooth complex hypersurface in a complex manifold and let v be continuous on a neighborhood of D. If v is plurisubharmonic off D, then it is plurisubharmonic everywhere on that neighborhood.*

Proof. It suffices to work in a polydisc with coordinates (z,w) in which $D = \{w = 0\}$. A restriction to an affine complex line not contained in D is subharmonic outside at most one point and is continuous at that point. Such a point is removable. One direct verification is to

take a small disc around it, let H be the harmonic function with the continuous boundary values of the restricted function, and apply the maximum principle on punctured annuli to $v - H + \varepsilon \log(|\zeta|/R)$. This expression is negative on the inner boundary for small enough inner radius, and is zero on the outer boundary. It is therefore nonpositive on the annulus. Let the inner radius tend to zero, then let ε decrease to zero. This proves the harmonic comparison inequality on the full disc, hence subharmonicity across its center.

For a complex line contained in D, translate that line by a sequence of small nonzero constants in the w coordinate. On each relatively compact disc the restrictions of v to the translated lines converge uniformly to its restriction to the original line, by continuity of v. The limit is subharmonic by Lemma 2.1. Thus the restriction to every affine complex line is subharmonic. The line-restriction characterization of a continuous plurisubharmonic function proves the assertion. This proof also shows that the removable extension takes the original continuous values on D. □

Lemma 2.3. *For every $\varepsilon > 0$ there exists a smooth convex function $\mathcal{M}_\varepsilon : \mathbb{R}^2 \to \mathbb{R}$ with the following properties:*

(i) *it is nondecreasing in each variable;*

(ii) $\mathcal{M}_\varepsilon(x + a, y + a) = a + \mathcal{M}_\varepsilon(x, y)$*;*

(iii) *it equals* $\max\{x, y\}$ *whenever* $|x - y| \geq \varepsilon$*;*

(iv) *if u, v are continuous plurisubharmonic functions, then $\mathcal{M}_\varepsilon(u, v)$ is plurisubharmonic;*

(v) *the maximum and the regularized maximum of two locally Lipschitz functions are locally Lipschitz.*

The same operations preserve θ-plurisubharmonicity when the two potentials are measured relative to the same smooth form θ.

Proof. Choose an even nonnegative smooth function χ supported in $(-\varepsilon, \varepsilon)$ with integral one and put

$$\rho_\varepsilon(t) = \int_{\mathbb{R}} |t - a| \chi(a)\, da, \qquad \mathcal{M}_\varepsilon(x, y) = \frac{x + y + \rho_\varepsilon(x - y)}{2}.$$

Convolution makes ρ_ε smooth. Convexity of the absolute value makes it convex, and $|\rho_\varepsilon'| \leq 1$. When $|t| \geq \varepsilon$, every $t - a$ has the sign of t, and evenness of χ gives $\int a\chi(a)\, da = 0$. Hence $\rho_\varepsilon(t) = |t|$ there. These facts prove (i)–(iii): the two first partial derivatives are $(1 + \rho_\varepsilon')/2$ and $(1 - \rho_\varepsilon')/2$, and the Hessian is a nonnegative multiple of $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$.

To prove (iv), restrict to a complex disc. The submean inequalities for u, v, monotonicity in each variable, and Jensen's inequality for the convex function $\mathcal{M}_\varepsilon$ give the submean inequality for $\mathcal{M}_\varepsilon(u, v)$. The function is continuous, so it is subharmonic on the disc. All disc restrictions have this property, which proves plurisubharmonicity. The proof for the ordinary maximum follows either from its submean inequality or by letting ε tend to zero locally uniformly.

For (v), the first derivatives of $\mathcal{M}_\varepsilon$ lie in $[0, 1]$ and sum to one. The fundamental theorem of calculus along a segment therefore gives

$$|\mathcal{M}_\varepsilon(x_1, y_1) - \mathcal{M}_\varepsilon(x_2, y_2)| \leq |x_1 - x_2| + |y_1 - y_2|.$$

The same inequality holds for the maximum. Composing this inequality with local Lipschitz bounds for u, v proves (v).

For the last assertion, add a local smooth potential φ_0 for θ. Translation invariance gives

$$\varphi_0 + \mathcal{M}_\varepsilon(u, v) = \mathcal{M}_\varepsilon(\varphi_0 + u, \varphi_0 + v).$$

The right-hand side is plurisubharmonic by (iv). The same identity holds for the ordinary maximum, proving the assertion on all frame charts. □

Remark 2.4. If the two inputs are smooth and strictly plurisubharmonic on a region, their regularized maximum is smooth and strictly plurisubharmonic there. Indeed, its complex Hessian is the sum of the two input Hessians with nonnegative coefficients adding to one, plus a positive semidefinite quadratic term from the Hessian of $\mathcal{M}_\varepsilon$. The weighted sum is positive definite at every point. The same conclusion holds for strictly θ-plurisubharmonic inputs after adding a local potential of θ.

2.3 Bounded potentials and minimal singularities

Definition 2.5. Let S be compact and θ a smooth closed real $(1,1)$-form. A function $u \in \mathrm{PSH}(S,\theta)$ has *minimal singularities* if for every $v \in \mathrm{PSH}(S,\theta)$, $v \not\equiv -\infty$, there exists a constant C_v such that $v \le u + C_v$ on S. The corresponding current $\theta + dd^c u$ is said to have minimal singularities.

This is the usual comparison of singularities; see [DPS01, Definition 1.4 and Theorem 1.5] and [BEGZ10, §1.4]. It does not require that u equal a specially normalized envelope.

Lemma 2.6. *Every bounded $u \in \mathrm{PSH}(S,\theta)$ on a compact connected manifold has minimal singularities. In particular every continuous such u has minimal singularities and defines a current with bounded local potentials.*

Proof. A θ-plurisubharmonic function $v \not\equiv -\infty$ is locally bounded above: add a local smooth potential of θ and use the local upper boundedness of a plurisubharmonic function. A finite relatively compact coordinate cover of the compact space gives $\sup_S v < \infty$. If u is bounded below, then for every point x,

$$v(x) \le \sup_S v \le u(x) + \sup_S v - \inf_S u.$$

This is the defining inequality with $C_v = \sup_S v - \inf_S u$. Continuity on the compact space implies boundedness. A local potential of the current is $\varphi_0 + u$; it is bounded on each relatively compact frame chart because φ_0 is smooth and u is bounded. □

3 The geometry of Grauert's example

We describe the construction with a positive line bundle of arbitrary degree. The original example in [Gra62, §4, paragraph 8(d)] uses a curve of genus two and a line bundle of degree one. Throughout this section, the projectivization of a vector bundle parametrizes *lines* in its fibers.

Assumption 3.1. Let B be a connected compact Riemann surface of genus $g \ge 2$. Let F be a holomorphic line bundle on B of degree $d > 0$, and choose a nonzero class

$$\xi \in H^1(B, F).$$

Here and below, $H^1(B,F)$ means cohomology of the sheaf of holomorphic sections of F.

The condition $H^1(B,F) \neq 0$ is an additional hypothesis; positivity of F alone does not imply it. The following choice provides the degree-one examples needed here.

Lemma 3.2. *For any $b_0 \in B$, the line bundle $F = \mathcal{O}_B(b_0)$ has degree one and*

$$h^1(B,F) = g - 1 > 0.$$

Proof. A section of $\mathcal{O}_B(b_0)$ is a meromorphic function on B with no poles except possibly a simple pole at b_0. If such a function were nonconstant, it would define a holomorphic map $B \to \mathbb{P}^1$ of degree one, because the degree equals the total pole order. A degree-one map between smooth compact curves is an isomorphism. This would contradict $g \geq 2$. Thus $h^0(B, F) = 1$. Riemann–Roch on B now gives

$$h^0(B, F) - h^1(B, F) = \deg F + 1 - g = 2 - g,$$

and the asserted formula follows. □

3.1 The affine bundle and its projective completion

Choose a sufficiently fine open cover $\{U_i\}$ of B on which F is trivial, with nowhere-vanishing holomorphic frames m_i. Write

$$m_j = g_{ij} m_i$$

on $U_i \cap U_j$, where the g_{ij} are nowhere-vanishing holomorphic functions. Thus fiber coordinates for the total space of F obey $t_i = g_{ij} t_j$. Represent ξ by a cocycle $\{a_{ij} m_i\}$, with a_{ij} holomorphic on $U_i \cap U_j$. The cocycle relations are

$$g_{ik} = g_{ij} g_{jk}, \qquad a_{ik} = a_{ij} + g_{ij} a_{jk}. \tag{3.1}$$

Glue the products $U_i \times \mathbb{C}$ by the affine transformations

$$t_i = g_{ij} t_j + a_{ij}. \tag{3.2}$$

The result is a holomorphic affine-line bundle $T \to B$. It is also called the F-torsor associated with ξ. The linear part of its transition functions is the one defining F, while the translation part records the cohomology class ξ.

Lemma 3.3. *The affine bundle $T \to B$ has no holomorphic section.*

Proof. A holomorphic section would be represented by holomorphic functions s_i on U_i satisfying

$$s_i = g_{ij} s_j + a_{ij}.$$

Equivalently,

$$a_{ij} m_i = s_i m_i - s_j m_j.$$

This expresses the cocycle representing ξ as a coboundary, contrary to $\xi \neq 0$. Conversely, a coboundary would give such a section. □

The affine transformations extend to automorphisms of $\mathbb{P}^1$ fixing the point at infinity. Adding this point to each fiber gives a smooth compact surface

$$q : X \longrightarrow B.$$

Its infinity section is denoted by C. Thus $C \cong B$ and $X \setminus C = T$. One can identify this completion with a projective bundle, including its holomorphic extension data, as follows.

Proposition 3.4. *The surface X is the projective bundle $\mathbb{P}_B(E)$ of lines in a rank-two vector bundle E fitting into the nonsplit exact sequence*

$$0 \longrightarrow F \longrightarrow E \longrightarrow \mathcal{O}_B \longrightarrow 0. \tag{3.3}$$

Its extension class is ξ, up to the harmless sign convention for identifying extension classes with Čech cocycles. The section C is $\mathbb{P}_B(F) \subset \mathbb{P}_B(E)$. In particular, X is a smooth projective surface.

Proof. Define E by identifying vector coordinates according to

$$\begin{pmatrix} u_i \\ v_i \end{pmatrix} = \begin{pmatrix} g_{ij} & a_{ij} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_j \\ v_j \end{pmatrix}.$$

The relations (3.1) imply the matrix cocycle condition. The first coordinate axis defines the subbundle F, and the second coordinate gives a quotient map $E \to \mathcal{O}_B$. A splitting would provide a lift of $1 \in \mathcal{O}_B$ with coordinates $(s_i, 1)$; its transition rule would be the one excluded in Lemma 3.3. Hence the extension is nonsplit.

On the chart $v_i \neq 0$ of the projectivized bundle, the coordinate $t_i = u_i/v_i$ satisfies (3.2). The complement of these charts is the section $v_i = 0$, namely $\mathbb{P}_B(F)$. This identifies the projectivization with the stated completion.

For projectivity, we use two standard algebraic facts explicitly. A compact Riemann surface is a smooth projective algebraic curve; and GAGA algebraizes every holomorphic vector bundle on a projective complex variety [Ser56]. Thus E is an algebraic vector bundle on the projective curve B. Its projective bundle is projective over B, and hence projective over $\mathbb{C}$ [Har77]. Smoothness follows already from its local description as $U_i \times \mathbb{P}^1$. □

3.2 The negative section and the distinguished line bundle

Near C, use the fiber coordinate $w_i = 1/t_i = v_i/u_i$. The affine transition rule becomes

$$w_i = \frac{w_j}{g_{ij} + a_{ij} w_j}, \qquad w_j = \frac{g_{ij} w_i}{1 - a_{ij} w_i}. \tag{3.4}$$

All these formulas are understood on sufficiently small overlap neighborhoods of C, where the denominators are nonzero.

Proposition 3.5. *The infinity section has normal bundle and self-intersection*

$$N_{C/X} \cong F^{-1}, \qquad C^2 = -d.$$

The distinguished line bundle

$$L_0 := \mathcal{O}_X(C) \otimes q^* F \tag{3.5}$$

restricts holomorphically trivially to C. With the lines convention for projectivization, $\mathcal{O}_X(C) \cong \mathcal{O}_{\mathbb{P}_B(E)}(1)$.

Proof. Let I_C be the ideal sheaf of C. The class of w_i generates I_C/I_C^2 on U_i. Modulo I_C^2, the first formula in (3.4) reads

$$w_i = g_{ij}^{-1} w_j.$$

This is the same transition rule as $m_i = g_{ij}^{-1} m_j$. Consequently $I_C/I_C^2 \cong F$ under the identification $C \cong B$. Dualizing gives $N_{C/X} \cong F^{-1}$. For a smooth divisor in a smooth surface, its self-intersection equals the degree of its normal bundle, so $C^2 = -d$.

The quotient $E \to \mathcal{O}_B$ in (3.3), composed with the tautological inclusion $\mathcal{O}_{\mathbb{P}_B(E)}(-1) \to q^*E$, is a section of $\mathcal{O}_{\mathbb{P}_B(E)}(1)$. Its zero divisor is exactly $\mathbb{P}_B(F) = C$, with multiplicity one. This proves the assertion about $\mathcal{O}_X(C)$. Finally, restriction of a divisor line bundle to its smooth divisor is the normal bundle. Therefore

$$L_0|_C \cong N_{C/X} \otimes F \cong F^{-1} \otimes F \cong \mathcal{O}_C.$$

□

For later use, the trivialization on C can be made completely explicit. Write s_C for the canonical section of $\mathcal{O}_X(C)$ and choose local frames σ_i near C with $s_C = w_i\sigma_i$. The frames

$$\ell_i := \sigma_i \otimes q^* m_i$$

of L_0 satisfy

$$\frac{\ell_i}{\ell_j} = \frac{w_j}{w_i}\, g_{ij}^{-1} = 1 + \frac{a_{ij}}{g_{ij}} w_j = \frac{1}{1 - a_{ij} w_i}. \tag{3.6}$$

In particular, the frames $\ell_i|_C$ agree on overlaps. The translation cocycle disappears upon restriction to C but persists in the first normal term of the transition functions.

3.3 Intersection numbers, nefness, and bigness

We use additive notation for tensor products when computing divisor classes. Let f denote a fiber of q. The relevant elementary intersection identities are

$$C^2 = -d, \quad C \cdot f = 1, \quad C \cdot q^*F = d, \quad (q^*F)^2 = 0, \quad q^*F \cdot f = 0. \tag{3.7}$$

Indeed, $q|_C$ is an isomorphism, the pullback of a line bundle from the base has degree zero on a fiber, and two classes pulled back from a curve have zero intersection on the surface. More explicitly, C meets each fiber transversely once, and q^*F is trivial on that fiber. Thus $L_0|_f \cong \mathcal{O}_{\mathbb{P}^1}(1)$.

Proposition 3.6. *The line bundle L_0 is nef and big. Its numerical invariants are*

$$L_0^2 = d, \qquad L_0 \cdot C = 0, \qquad L_0 \cdot f = 1.$$

Moreover, $L_0 \cdot D > 0$ for every irreducible curve $D \neq C$. The line bundle

$$A := L_0^{\otimes 2} \otimes \mathcal{O}_X(-C) = \mathcal{O}_X(C) \otimes q^*F^2 \tag{3.8}$$

is ample and satisfies $A^2 = 3d$ and $A \cdot C = d$.

Proof. The displayed invariants follow from $L_0 = C + q^*F$ and (3.7). To check every curve, let $D \neq C$ be irreducible. The local intersection multiplicities of two distinct curves on a smooth surface are nonnegative, so $C \cdot D \geq 0$.

If $q(D)$ is a point, then D is a fiber, since each fiber is an irreducible copy of $\mathbb{P}^1$. In this case $L_0 \cdot D = 1$. Otherwise, the induced map from the normalization of D to B has a positive degree n. Pullback of a degree-d line bundle has degree nd, and hence

$$L_0 \cdot D = C \cdot D + nd > 0.$$

Together with $L_0 \cdot C = 0$, this proves nefness, using the intersection characterization of nef line bundles on projective varieties. It also proves that the null locus, defined here as the union of curves having zero intersection with L_0, is exactly C.

For $A = C + 2q^*F$, the same calculation gives

$$A^2 = 3d > 0, \qquad A \cdot C = d > 0, \qquad A \cdot f = 1.$$

For a horizontal irreducible curve $D \neq C$ of degree n over B, one has $A \cdot D = C \cdot D + 2nd > 0$. We now apply the Nakai–Moishezon criterion in its surface form: a line bundle on a projective surface is ample if its square is positive and its intersection with every irreducible curve is positive [Har77, Laz04]. The hypotheses have all been checked, so A is ample.

For completeness, bigness can be verified directly from growth of sections, without appealing to the general nef-and-positive-square criterion. Multiplication by s_C^k gives an injection

$$H^0(X, A^{\otimes k}) \longrightarrow H^0(X, L_0^{\otimes 2k}),$$

because $A^{\otimes k} \otimes \mathcal{O}_X(kC) = L_0^{\otimes 2k}$ and s_C does not vanish identically on any open set. The Hilbert polynomial and Serre vanishing for an ample line bundle on a projective surface give

$$h^0(X, A^{\otimes k}) = \frac{A^2}{2}k^2 + O(k) = \frac{3d}{2}k^2 + O(k).$$

Consequently

$$\limsup_{m\to\infty} \frac{h^0(X, L_0^{\otimes m})}{m^2} \geq \frac{3d}{8} > 0.$$

This is the section-growth definition of a big line bundle. □

Remark 3.7. The formula in [FT18, discussion preceding Problem 2.2] can be matched with an actual line bundle, not merely a numerical class. Taking the ample bundle A of (3.8), one has, in additive notation,

$$A - \frac{A \cdot C}{C^2}C = A + C = 2L_0.$$

Thus the displayed construction in that paper includes $L_0^{\otimes 2}$. The bundle L_0 itself is a convenient distinguished choice on the same numerical ray.

3.4 The trace argument and failure of semiampleness

Lemma 3.8. *The affine bundle $T = X \setminus C$ contains no compact complex analytic curve.*

Proof. Suppose there were such a curve. Passing to the reduced structure and then to an irreducible component, choose a compact irreducible curve $D \subset T$. Let $\nu : \widetilde{D} \to D$ be its normalization. The resulting map

$$p := q \circ \nu : \widetilde{D} \longrightarrow B$$

cannot be constant. Otherwise D would lie in one affine fiber $q^{-1}(b) \cap T \cong \mathbb{C}$; its affine coordinate would pull back to a holomorphic function on the compact connected Riemann surface $\widetilde{D}$, hence would be constant, forcing D to be a point.

A nonconstant holomorphic map between compact connected Riemann surfaces is surjective and finite. Let its degree be $n \geq 1$. Away from the finitely many branch values of p, define on U_i

$$s_i(b) = \frac{1}{n} \sum_{x \in p^{-1}(b)} t_i(\nu(x)). \tag{3.9}$$

There are n points in this sum over an unbranched fiber. Locally they vary holomorphically. Permuting the sheets does not change their sum, so s_i is a single-valued holomorphic function away from the branch values.

We spell out why branching causes no problem. Around a branch value $b_1 \in U_i$, choose a small closed disk contained in U_i. Its inverse image in $\widetilde{D}$ is compact. The holomorphic function $t_i \circ \nu$ is bounded there, so the averages in (3.9) remain bounded as b approaches b_1. The removable-singularity theorem extends s_i holomorphically across b_1. This proves that s_i is holomorphic on all of U_i.

On an unbranched fiber over $U_i \cap U_j$, summing (3.2) over its n points gives

$$s_i = g_{ij}s_j + a_{ij}.$$

The identity extends across branch values by holomorphicity. Thus the s_i define a holomorphic section of $T \to B$, contradicting Lemma 3.3. □

Proposition 3.9. *For every integer $m > 0$, every global section of $L_0^{\otimes m}$ vanishes identically on C. In particular, L_0 is not semiample. Its stable base locus is exactly C.*

Proof. Since $L_0|_C \cong \mathcal{O}_C$, the restriction of a section $s \in H^0(X, L_0^{\otimes m})$ is a constant in the fixed trivialization. If this constant were nonzero, s would be nowhere zero on C. Its zero divisor would therefore be disjoint from C.

That zero divisor is nonempty. Indeed, every fiber meets C in one point, so the restriction of s to any fiber would be a nonzero section of $\mathcal{O}_{\mathbb{P}^1}(m)$. Such a section has zeros because $m > 0$. A nonzero holomorphic section of a line bundle on a smooth surface has a divisor of zeros; here this is a nonempty compact curve in $X \setminus C$. This contradicts Lemma 3.8. Hence $s|_C = 0$ for every s.

No positive tensor power of L_0 is globally generated, which is precisely the failure of semiampleness. On the other hand, $A^{\otimes k}$ is globally generated for all sufficiently large k, since A is ample. For each $x \notin C$, choose a section of $A^{\otimes k}$ nonzero at x and multiply it by s_C^k. The resulting section of $L_0^{\otimes 2k}$ is nonzero at x. Thus no point outside C belongs to the stable base locus, while the first part shows that every point of C does. □

3.5 The canonical bundle and the contraction

Proposition 3.10. *The canonical bundle of X is*

$$K_X \cong \mathcal{O}_X(-2C) \otimes q^*(K_B \otimes F^{-1}). \tag{3.10}$$

In particular,

$$K_X \cdot C = 2g - 2 + d > 0,$$

and $H^0(X, K_X^{\otimes m}) = 0$ for every $m > 0$. The smooth surface X therefore has Kodaira dimension $-\infty$.

Proof. Let $S = \mathcal{O}_{\mathbb{P}_B(E)}(-1)$ be the tautological subbundle of q^*E. The quotient $Q = q^*E/S$ has rank one. Since $\det E = F$, taking determinants gives $Q \cong q^*F \otimes S^{-1}$. The relative tangent bundle of a projective bundle parametrizes first-order deformations of a line in its fixed ambient fiber. At a line $\ell \subset E_b$, a first-order deformation is the graph of a linear map $\ell \to E_b/\ell$. These identifications vary holomorphically, giving

$$T_{X/B} \cong \operatorname{Hom}(S, Q) \cong \mathcal{O}_X(2C) \otimes q^*F.$$

Dualizing and tensoring with q^*K_B proves (3.10). Intersecting with C gives $2d + (2g-2-d) = 2g - 2 + d$, also in agreement with adjunction.

On every fiber $f \cong \mathbb{P}^1$, one has $K_X^{\otimes m}|_f \cong \mathcal{O}_{\mathbb{P}^1}(-2m)$, which has no nonzero holomorphic sections. Every global pluricanonical section therefore vanishes on every fiber, hence on all of X. □

The next existence statement uses Grauert's analytic contraction theorem as a classical background result, rather than an algebraic base-point-free theorem. In the local form needed here, that theorem says that a connected compact curve in a smooth complex surface whose intersection matrix is negative definite has a neighborhood admitting a proper holomorphic contraction onto a normal complex surface. The curve is mapped to one point and the map is biholomorphic off the curve [Gra62, BHPV04].

Proposition 3.11. *There is a proper holomorphic map*

$$\pi : X \longrightarrow Y$$

onto a normal compact complex surface such that $\pi(C) = \{y\}$ and π is biholomorphic from $X \setminus C$ onto $Y \setminus \{y\}$. The point y is singular, and it is the only singular point of Y. The surface Y is not projective.

Proof. The intersection matrix of the single curve C is the negative one-by-one matrix $(-d)$, so Grauert's theorem applies. Let $\pi_U : U \to V$ be its local contraction, with V normal and $\pi_U(C) = \{y\}$. Glue V to $X \setminus C$ using the biholomorphism $U \setminus C \cong V \setminus \{y\}$. Topologically the resulting space is the quotient obtained by collapsing the compact closed subset C to a point. The proper local contraction induces the quotient topology near that point. This quotient is Hausdorff and compact, and its two indicated charts supply a complex-space structure. Normality holds near y by the local theorem and elsewhere because $X \setminus C$ is smooth. The induced map π is holomorphic and proper; properness also follows directly from compactness of X and the Hausdorff property of Y.

We verify that y cannot be smooth. If it were, choose local coordinates z_1, z_2 near y. The pulled-back two-form

$$\eta = \pi^*(dz_1 \wedge dz_2)$$

would be a holomorphic section of K_X over the inverse image of that coordinate neighborhood. It is nowhere zero off C, since π is biholomorphic there. It vanishes on C, because $\pi|_C$ is constant and therefore $d\pi$ annihilates its tangent direction. Its divisor is consequently aC for an integer $a \geq 1$. Restricting the corresponding line-bundle identity to C gives

$$K_X \cdot C = aC^2 = -ad < 0.$$

This contradicts Proposition 3.10, or directly the adjunction identity $K_X \cdot C = 2g - 2 + d$. Thus y is singular. All other points are smooth by the stated biholomorphism.

Finally, suppose Y were projective. In a projective embedding, choose a hyperplane that avoids y and does not contain Y. Its intersection with the projective surface Y is a nonempty compact curve, disjoint from y. The inverse image of that curve under π is a compact curve in $X \setminus C = T$, contradicting Lemma 3.8. Therefore Y is not projective. □

Remark 3.12. The surface Y is bimeromorphic to the projective surface X, and hence is Moishezon. The preceding argument establishes its nonprojectivity directly. We do not infer non-Kählerness from a general assertion about singular Moishezon spaces. A separate argument using the metric obstruction will establish that property later. Likewise, the Kodaira-dimension statement above concerns the smooth ruled surface X; no assertion about reflexive powers of the canonical sheaf on the singular space Y is needed here.

4 The obstruction to continuously differentiable potentials

We now prove that the regularity of the metric constructed in Section 5 cannot be improved to C^1. The argument is local near the infinity section, although the compactness of that section is essential. Its analytic ingredient is a first-derivative property of plurisubharmonic functions. In particular, we will not differentiate a C^1 weight twice or assume that its curvature has continuous coefficients.

4.1 Regularity and the first normal derivative

For a positive Hermitian metric h on a holomorphic line bundle, write

$$|e|_h^2 = e^{-\varphi}$$

in a local holomorphic frame e. We say that h is of class C^1 if its local weights φ are of class C^1 as real-valued functions. If $e' = Ge$, where G is a nowhere-vanishing holomorphic function, then

$$\varphi' = \varphi - \log |G|^2.$$

The change of weight is smooth, so this definition is independent of the holomorphic frame. The regularity of the metric coefficients is also unchanged under a smooth change of frame. Plurisubharmonicity of the weights, however, is asserted in holomorphic frames. A C^1 semipositive metric means that these weights are plurisubharmonic, or equivalently that their distributional dd^c is positive.

We use the Wirtinger derivatives

$$\partial_z = \frac{1}{2}(\partial_x - \sqrt{-1}\partial_y), \qquad \partial_{\bar{z}} = \frac{1}{2}(\partial_x + \sqrt{-1}\partial_y), \qquad \Delta = 4\partial_z\partial_{\bar{z}}.$$

Lemma 4.1 (Holomorphicity of the first normal derivative)**.** *Let $D \subset \mathbb{C}$ be a disc, let $\Omega \subset \mathbb{C}^2$ be an open neighborhood of $D \times \{0\}$, and let $\varphi \in C^1(\Omega) \cap \mathrm{PSH}(\Omega)$ be real-valued. Suppose that $\varphi(z,0)$ is constant on D. Then*

$$b(z) := \partial_w \varphi(z,0)$$

is holomorphic on D.

Proof. Subtracting a constant, we may assume that $\varphi(z,0) = 0$. The function b is initially only known to be continuous. We first record carefully the uniformity of the first-order expansion. Let $K \subset D$ be compact and choose $\rho > 0$ such that a neighborhood of $K \times \{|w| \le \rho\}$ is contained in Ω. The fundamental theorem of calculus, applied to the real parameter s, gives

$$\begin{aligned}\varphi(z,w) &= \int_0^1 \frac{d}{ds}\varphi(z,sw)\,ds \\ &= 2\,\mathrm{Re}\left(w\int_0^1 \partial_w\varphi(z,sw)\,ds\right).\end{aligned}$$

Here we used that φ is real-valued. Consequently

$$\varphi(z,w) = 2\,\mathrm{Re}\big(b(z)w\big) + R(z,w), \qquad \sup_{z\in K}|R(z,w)| \le 2|w|\omega_K(|w|), \tag{4.1}$$

where one may take

$$\omega_K(\varepsilon) = \sup_{\substack{z\in K\\ |v|\le\varepsilon}} \big|\partial_w\varphi(z,v) - \partial_w\varphi(z,0)\big|.$$

Continuity of the first derivative on a compact neighborhood implies $\omega_K(\varepsilon) \longrightarrow 0$ as $\varepsilon \downarrow 0$. Thus the remainder is $o(|w|)$ locally uniformly in the base variable.

Let f be holomorphic on D, and work first on a relatively compact subdisc $D' \Subset D$. For all sufficiently small real $t > 0$, the two maps

$$z \longmapsto (z, tf(z)), \qquad z \longmapsto (z, -tf(z))$$

take a neighborhood of $\overline{D'}$ into Ω and are holomorphic. The functions

$$u_t^+(z) = \frac{\varphi(z,tf(z))}{t}, \qquad u_t^-(z) = \frac{\varphi(z,-tf(z))}{t}$$

are therefore subharmonic on that neighborhood. The denominator is positive in both expressions; no multiplication of a subharmonic function by a negative constant is being used. Estimate (4.1) proves the locally uniform limits

$$u_t^+ \longrightarrow 2\,\mathrm{Re}(bf), \qquad u_t^- \longrightarrow -2\,\mathrm{Re}(bf). \tag{4.2}$$

For example, on a compact subset where $|f| \le A$, the error in either limit is bounded by $2A\omega_K(tA)$.

For completeness, these limits are subharmonic directly from the submean inequality. If a closed disc lies in D', the inequality for $u_t^{\pm}$ compares its value at the center with its circular mean. Uniform convergence on that disc and its boundary permits passage to the limit. The resulting functions are continuous and satisfy the same inequality on every such disc, hence are subharmonic. It follows from (4.2) that both $\mathrm{Re}(bf)$ and its negative are subharmonic.

We also recall why this implies ordinary harmonicity, with no regularity assumption beyond continuity. Applying the submean inequality to a continuous function and to its negative gives equality with every sufficiently small circular mean. Convolution with a smooth radial mollifier is a weighted average of these circular means and therefore equals the function wherever the convolution is defined. The function is consequently smooth there. Its circular-mean identity, expanded to second order, then gives $\Delta h = 0$. Thus $\mathrm{Re}(bf)$ is harmonic on D', and hence on D because D' was arbitrary.

Take successively $f = 1$, $f = \sqrt{-1}$, $f = z$, and $f = \sqrt{-1}z$. The first two choices show that the real and imaginary parts of b are harmonic; the last two show the same for zb. In particular b is smooth and

$$\Delta b = 0, \qquad \Delta(zb) = 0.$$

The product rule for Wirtinger derivatives gives the exact identity

$$\Delta(zb) = 4\partial_z\big(z\partial_{\bar z} b\big) = 4\partial_{\bar z} b + z\Delta b.$$

Hence $\partial_{\bar z} b = 0$, which proves that b is holomorphic. □

Remark 4.2. There is a shorter proof when φ is C^2. Its complex Hessian at $(z, 0)$ is a positive semidefinite Hermitian matrix, and its $z\bar z$ entry vanishes because $\varphi(z, 0) = 0$. Positivity of a two-by-two Hermitian matrix with one zero diagonal entry forces the corresponding mixed entries to vanish. Thus $\varphi_{w\bar z}(z, 0) = 0$, and $\partial_{\bar z} b = 0$. Lemma 4.1 replaces this Hessian argument when only one continuous derivative is available.

4.2 The transition calculation on the infinity section

We spell out the conventions needed for the application. Let $q : X \longrightarrow B$ be the ruled surface obtained by compactifying the affine line bundle associated with a nonzero class $\xi \in H^1(B, F)$, and let $C \subset X$ be its infinity section. Choose local holomorphic frames m_i of F and affine fiber coordinates t_i over open sets $U_i \subset B$, with transition formulas

$$m_j = g_{ij} m_i, \qquad t_i = g_{ij} t_j + a_{ij}. \tag{4.3}$$

The functions g_{ij} are nowhere vanishing and holomorphic, the functions a_{ij} are holomorphic, and the cocycle $\{a_{ij} m_i\}$ represents ξ. Base points are suppressed in these formulas; a function on U_j is understood to be restricted to the overlap and expressed in the chosen base coordinate there.

A holomorphic section of the affine line bundle would be given by holomorphic functions s_i satisfying

$$s_i = g_{ij} s_j + a_{ij}. \tag{4.4}$$

Multiplying this identity by m_i gives $s_i m_i - s_j m_j = a_{ij} m_i$. Thus such functions would express the cocycle as a coboundary. Since $\xi \neq 0$, no such section exists.

Put

$$L_0 = \mathcal{O}_X(C) \otimes q^* F.$$

Near C, use $w_i = 1/t_i$ as the transverse coordinate, so that $C = \{w_i = 0\}$. Formula (4.3) becomes

$$w_i = \frac{w_j}{g_{ij} + a_{ij} w_j}, \qquad w_j = \frac{g_{ij} w_i}{1 - a_{ij} w_i}. \tag{4.5}$$

All expressions here are considered on sufficiently small overlapping neighborhoods of C, where their denominators do not vanish. If σ_C is the canonical section of $\mathcal{O}_X(C)$, choose its natural local frames v_i by

$$\sigma_C = w_i v_i.$$

The identity $w_i v_i = w_j v_j$ and (4.5) imply

$$v_j = \frac{1 - a_{ij} w_i}{g_{ij}} v_i.$$

Consequently the holomorphic frames $e_i = q^* m_i \otimes v_i$ of L_0 satisfy the exact formula

$$e_j = (1 - a_{ij} w_i) e_i. \tag{4.6}$$

In particular, their restrictions agree on C and define a global nowhere-vanishing holomorphic section of $L_0|_C$.

Theorem 4.3 (Nonexistence of a C^1 semipositive metric)**.** *Assume that the affine extension class $\xi \in H^1(B, F)$ is nonzero. Then L_0 admits no C^1 Hermitian metric with semipositive curvature current on any open neighborhood of C in X.*

Proof. Suppose that such a metric h exists on a neighborhood N of C. Shrink the local coordinate neighborhoods used above so that they lie in N, and write

$$|e_i|_h^2 = e^{-\varphi_i}.$$

The functions φ_i are C^1 and plurisubharmonic. The exact transformation law (4.6) gives

$$\varphi_j(z, w_j) = \varphi_i(z, w_i) - \log|1 - a_{ij}(z) w_i|^2, \qquad w_j = \frac{g_{ij}(z) w_i}{1 - a_{ij}(z) w_i}. \tag{4.7}$$

Here the same notation z denotes the base point on the overlap; any change between base coordinate charts depends only on that base point.

At $w_i = w_j = 0$, the logarithmic term vanishes. Thus the restrictions $\varphi_i|_C$ glue to a global continuous real-valued function on C. The restriction of a finite plurisubharmonic function to a holomorphic curve is subharmonic. Since C is compact and connected, the maximum principle implies that this global function is constant. Subtracting the same constant from all weights, we arrange that

$$\varphi_i(z, 0) = 0. \tag{4.8}$$

This normalization does not change curvature or regularity.

Set

$$b_i(z) = \partial_{w_i} \varphi_i(z, 0).$$

Lemma 4.1, applied on coordinate discs in U_i, proves that every b_i is holomorphic. We may differentiate (4.7) once in w_i, because the weights are C^1. At $w_i = 0$, the derivative of w_j with respect to w_i is g_{ij}, and its antiholomorphic counterpart contributes nothing to the Wirtinger derivative. Moreover,

$$\partial_{w_i} \log|1 - a_{ij} w_i|^2 \Big|_{w_i = 0} = -a_{ij}.$$

We therefore obtain

$$g_{ij} b_j = b_i + a_{ij}, \qquad b_i = g_{ij} b_j - a_{ij}. \tag{4.9}$$

Equivalently, one can substitute the locally uniform expansions from (4.1) into (4.7) and compare the real linear terms in w_i; this gives the same identity.

Now put $s_i = -b_i$. The functions s_i are holomorphic, and (4.9) becomes precisely

$$s_i = g_{ij} s_j + a_{ij}.$$

They define the forbidden affine section from (4.4). Explicitly,

$$b_j m_j - b_i m_i = a_{ij} m_i$$

expresses the cocycle representing ξ as a coboundary. This contradicts $\xi \neq 0$ and proves the theorem. □

Corollary 4.4. *For every integer $m \geq 1$, the line bundle $L_0^{\otimes m}$ admits no C^1 semipositive Hermitian metric on a neighborhood of C. Equivalently, relative to any smooth reference metric on $L_0^{\otimes m}$, no semipositive metric potential can be C^1 on a neighborhood of C.*

Proof. A positive metric on $L_0^{\otimes m}$ has a unique positive mth root as a metric on the already given line bundle L_0. In the frames $e_i^{\otimes m}$ and e_i, this simply divides every local weight by m. It preserves C^1 regularity and plurisubharmonicity, contradicting Theorem 4.3. Subtracting the smooth local weight of a reference metric does not change whether a potential is C^1. □

4.3 Relation with the first obstruction class

Remark 4.5. The calculation also identifies the first obstruction in the transition-function approach. With the convention that this obstruction is represented by the coefficient of w_i in e_j/e_i, formula (4.6) gives

$$u_1(C, X, L_0) = [\{-a_{ij}\, dw_i\}] \in H^1(C, N_{C/X}^{-1}).$$

Along C, formula (4.5) yields $dw_j = g_{ij} dw_i$. Hence the identification $N_{C/X}^{-1} \simeq F$ sending dw_i to m_i identifies this class with $-\xi$. Reversing the convention for the transition cocycle changes the sign but not its nonvanishing. For a tensor power, the expansion

$$\frac{e_j^{\otimes m}}{e_i^{\otimes m}} = (1 - a_{ij} w_i)^m = 1 - m a_{ij} w_i + O(w_i^2)$$

identifies the corresponding class with $-m\xi$.

Koike's line-bundle obstruction theory [Koi21] extends the transition-function viewpoint originating in Ueda's neighborhood theory [Ued83]. Zhang computes the nonzero first obstruction for the original Grauert construction and applies the smooth semipositivity obstruction [Zha25]. The proof above is given independently: its additional analytic ingredient is Lemma 4.1, and it establishes nonexistence for C^1 weights without invoking a theorem that assumes a smooth metric.

Remark 4.6 (What the regularity conclusion means). The obstruction is to continuous first derivatives, not to bounded or Lipschitz local weights. A Lipschitz function need not possess the uniform real-linear first-order expansion (4.1) along C. In particular, the transverse model $|w|$ is Lipschitz but is not differentiable at $w = 0$. This is exactly the type of regularity permitted by the norm-of-section potential in the construction. Thus Lipschitz existence and Theorem 4.3 are compatible. Together they give the precise endpoint statement used here: Lipschitz semipositive potentials exist, whereas C^1 semipositive potentials do not. No classification of other, weaker differentiability notions is asserted.

5 Constructing a Lipschitz semipositive metric

The construction in this section applies more generally than to a ruled surface. Its local input is a negative smooth curve and a line bundle whose restriction to that curve is flat. The global input is an exact ample twist by the curve. We prove the local estimates and the gluing separately, using the same smooth reference metric throughout.

5.1 Reference metrics near a negative curve

Let S be a smooth complex surface and let $C \subset S$ be a smooth, compact, connected complex curve with $C^2 < 0$. Let L be a holomorphic line bundle on a neighborhood of C. Assume that $L|_C$ has a smooth Hermitian metric of zero curvature. In our application $L|_C$ is holomorphically trivial, so the constant metric in a global nonvanishing frame has this property.

We explain two choices of smooth metrics. First choose a smooth metric h_0 on L whose restriction is the specified flat metric. To do this, start with any smooth metric h_0'. The logarithm of the ratio of the desired metric to $h_0'|_C$ is a smooth real function on C. Extend that function smoothly to a neighborhood of C, and multiply h_0' by its exponential. Smooth extension is available from a smooth tubular neighborhood, or from smooth coordinate extensions and a partition of unity. No holomorphic tubular neighborhood is used. With

$$\theta = c_1(L, h_0),$$

we have $\theta|_{TC} = 0$.

Second, choose a smooth Hermitian metric k on $\mathcal{O}_S(C)$ such that

$$\eta := -c_1(\mathcal{O}_S(C), k) \quad \text{is strictly positive on } TC. \tag{5.1}$$

Here is why such a choice is possible. The restriction $\mathcal{O}_S(C)|_C = N_{C/S}$ has degree $C^2 < 0$. On a compact Riemann surface, a line bundle of negative degree has a smooth metric with strictly negative curvature. In detail, fix a positive area form ω_C of integral one and any metric with curvature γ. The form $C^2\omega_C - \gamma$ has integral zero. The Poisson equation on the compact curve gives a smooth real function v with $dd^c v = C^2\omega_C - \gamma$. Multiplying the metric by e^{-v} gives curvature $C^2\omega_C$. The Poisson solvability used here is the usual compact-surface statement: the Laplacian has constants as kernel and has as range the smooth functions of integral zero [Dem12b]. Extend this metric from C to $\mathcal{O}_S(C)$ by the smooth ratio construction just described. Its negative curvature has the required positive restriction.

Let s_C be the canonical section of $\mathcal{O}_S(C)$ and set

$$r = |s_C|_k^2, \qquad s = r^{1/2}. \tag{5.2}$$

Thus r is a smooth nonnegative function, with zero set C. In a holomorphic coordinate chart (z, w) with $C = \{w = 0\}$, choose a frame v of $\mathcal{O}_S(C)$ such that $s_C = wv$ and put $|v|_k^2 = e^\lambda$. Then

$$r = |w|^2 e^\lambda, \qquad s = |w| e^{\lambda/2}, \qquad \eta = dd^c\lambda. \tag{5.3}$$

In particular, s is locally Lipschitz, including on C, and is smooth off C. Indeed, $w \mapsto |w|$ is Lipschitz, and $e^{\lambda/2}$ is smooth with bounded first derivatives on every relatively compact coordinate chart; their product is Lipschitz there.

On $S \setminus C$ one has

$$dd^c \log r = \eta. \tag{5.4}$$

This is an identity on the complement only. As currents across C, the correct identity is

$$dd^c \log r = [C] + \eta. \tag{5.5}$$

It follows from $dd^c \log |w|^2 = [w = 0]$ with our normalization. This distinction will be essential for the global auxiliary metric.

5.2 A frame adapted to the level sets of the norm

We will estimate Hermitian coefficient matrices as in Section 2: a real $(1,1)$-form is written as $(\sqrt{-1}/2\pi)\sum T_{\alpha\bar{\beta}}dz_\alpha \wedge d\bar{z}_\beta$, and positivity is tested on $(T_{\alpha\bar{\beta}})$. All coefficients below use this convention.

Shrink a relatively compact coordinate chart until $|w\lambda_w| \le 1/2$. Define two smooth $(1,0)$ vector fields

$$H = \partial_z - \frac{w\lambda_z}{1+w\lambda_w}\partial_w, \qquad V = \partial_w. \tag{5.6}$$

They form a frame because the coefficient of ∂_z in H is one. Its change-of-frame matrix is triangular with determinant one. It extends smoothly to $w = 0$, where $H = \partial_z$. On $w \neq 0$, direct substitution gives

$$\partial \log r(H) = \lambda_z - \frac{w\lambda_z}{1+w\lambda_w}\left(\frac{1}{w} + \lambda_w\right) = 0, \tag{5.7}$$

$$\partial \log r(V) = \frac{1}{w} + \lambda_w. \tag{5.8}$$

The vector fields need not be holomorphic. We use them only to *evaluate* tensors computed in the holomorphic coordinates. In particular, none of the formulas below replaces a Hessian coefficient by a repeated derivative along the nonholomorphic field H.

For $\delta > 0$, differentiating $r^\delta = e^{\delta \log r}$ on the complement of C gives the exact formula

$$dd^c r^\delta = \delta r^\delta \eta + \delta^2 r^\delta \frac{\sqrt{-1}}{2\pi}\partial \log r \wedge \bar{\partial} \log r. \tag{5.9}$$

In the frame (H, V) this reads

$$(dd^c r^\delta)_{H\bar{H}} = \delta r^\delta \eta_{H\bar{H}}, \tag{5.10}$$

$$(dd^c r^\delta)_{H\bar{V}} = \delta r^\delta \eta_{H\bar{V}}, \tag{5.11}$$

$$(dd^c r^\delta)_{V\bar{V}} = \delta r^\delta \eta_{V\bar{V}} + \delta^2 r^\delta \left|\frac{1}{w} + \lambda_w\right|^2. \tag{5.12}$$

The cancellation in (5.11) is why we use the adapted frame. Estimating entries independently in the original frame would hide the useful rank-one structure of the second term in (5.9).

Lemma 5.1 (Uniform estimates for the square-root cusp)**.** *There are a neighborhood U of C, finitely many charts covering U, and constants $a, b, K > 0$ with the following properties on $U \setminus C$. In each chart, using the frame (5.6),*

$$(dd^c s)_{H\bar{H}} \ge as, \qquad (dd^c s)_{V\bar{V}} \ge \frac{b}{s}, \qquad |(dd^c s)_{H\bar{V}}| \le Ks, \tag{5.13}$$

$$\theta_{H\bar{H}} \ge -Ks, \qquad \theta_{V\bar{V}} \ge -K, \qquad |\theta_{H\bar{V}}| \le K. \tag{5.14}$$

All constants are independent of the point and of the chart in this finite covering.

Proof. Choose finitely many relatively compact charts around C and smaller charts whose intersections with C still cover it. Work on these smaller charts, so all smooth functions and their needed derivatives have bounds supplied by the larger charts. After shrinking in the normal direction, $|w\lambda_w| \le 1/2$ holds on every chart. Since $H|_C = \partial_z$ and $\eta|_{TC} > 0$, continuity and the finite cover give a number $\mu > 0$ such that $\eta_{H\bar{H}} \ge \mu$ on a common smaller neighborhood. There is also a common $A \ge 1$ such that

$$|\eta_{H\bar{V}}| \le A, \qquad |\eta_{V\bar{V}}| \le A, \qquad A^{-1} \le e^\lambda \le A.$$

Put $\delta = 1/2$ in (5.10)–(5.12). The $H\bar{H}$ and $H\bar{V}$ formulas immediately give

$$(dd^c s)_{H\bar{H}} \geq \frac{\mu}{2} s, \qquad |(dd^c s)_{H\bar{V}}| \leq \frac{A}{2} s.$$

The bound $|w\lambda_w| \leq 1/2$ implies

$$\left|\frac{1}{w} + \lambda_w\right| = \frac{|1 + w\lambda_w|}{|w|} \geq \frac{1}{2|w|}.$$

Using $s^2 = |w|^2 e^{\lambda}$, we obtain

$$\begin{aligned}(dd^c s)_{V\bar{V}} &\geq -\frac{A}{2} s + \frac{s}{16|w|^2} = -\frac{A}{2} s + \frac{e^{\lambda}}{16 s} \\ &\geq -\frac{A}{2} s + \frac{1}{16 A s}.\end{aligned}$$

Shrink the neighborhood so that $s^2 \leq 1/(16A^2)$. Then $(A/2)s \leq 1/(32As)$, and the last expression is at least $1/(32As)$. We may take $a = \mu/2$ and $b = 1/(32A)$.

For the reference form, $\theta_{H\bar{H}}$ is a smooth real function on the entire chart, including $w = 0$. Its value there is zero because $H|_C = \partial_z$ and $\theta|_{TC} = 0$. The real mean-value estimate in the two normal variables therefore gives $|\theta_{H\bar{H}}(z, w)| \leq A_1 |w|$ for a uniform constant A_1. For example, integrate its first normal derivative along the segment $t \mapsto (z, tw)$, $0 \leq t \leq 1$, in a product subchart. Since $|w| \leq A^{1/2} s$, this is at most $A_1 A^{1/2} s$. The other two coefficients of θ in the smooth frame are uniformly bounded. Enlarge K to dominate those bounds, $A_1 A^{1/2}$, and $A/2$. The displayed inequalities follow on the common neighborhood. □

5.3 Fixing the coefficient before shrinking the neighborhood

Theorem 5.2 (Local Lipschitz existence)**.** *Let C and L satisfy the assumptions at the beginning of this section. There exist a neighborhood U of C and $M > 0$ such that*

$$T_M = \theta + M dd^c s \tag{5.15}$$

is a positive current on U and a smooth strictly positive $(1,1)$-form on $U \setminus C$. The metric $h_{\mathrm{loc}} = h_0 e^{-Ms}$ is positive definite with locally Lipschitz weights on U.

Proof. Choose a, b, K as in Lemma 5.1. First choose one constant $M > 0$ satisfying

$$Ma - K > 0, \qquad (Ma - K)\frac{Mb}{2} > (K+1)^2. \tag{5.16}$$

Such a choice exists: the left side of the second inequality is a quadratic polynomial in M with positive leading coefficient $ab/2$. This choice is made before any further shrinking.

With this M fixed, shrink U until

$$Ks \leq \frac{Mb}{2}, \qquad MKs \leq 1 \tag{5.17}$$

throughout U. This is possible because s is continuous and zero on the compact set C. The estimates in the preceding lemma now give

$$\begin{aligned}(T_M)_{H\bar{H}} &\geq (Ma - K)s, \\ (T_M)_{V\bar{V}} &\geq \frac{Mb}{s} - K \geq \frac{Mb}{2s}, \\ |(T_M)_{H\bar{V}}| &\leq K + MKs \leq K + 1.\end{aligned}$$

Both diagonal coefficients are strictly positive for $s > 0$. Moreover,

$$\begin{aligned}\det(T_M)_{(H,V)} &= (T_M)_{H\bar{H}}(T_M)_{V\bar{V}} - |(T_M)_{H\bar{V}}|^2 \\ &\geq (Ma - K)\frac{Mb}{2} - (K+1)^2 > 0.\end{aligned}$$

The two-by-two Hermitian matrix is therefore positive definite. Because (H, V) is a frame, this is positivity of the form itself. This proves strict positivity on $U \setminus C$.

Let e be any local holomorphic frame of L and write $|e|^2_{h_0} = e^{-\varphi_0}$. The weight $\varphi_0 + Ms$ is smooth and strictly plurisubharmonic off C and is continuous across C. The removable lemma Lemma 2.2 makes it plurisubharmonic on the whole chart. Its distributional dd^c is thus positive on that chart, including on C. These local statements give positivity of T_M as a current on U. Finally, φ_0 is smooth and s is locally Lipschitz by (5.3), so the weight has the asserted regularity. It is finite everywhere; its exponential defines a positive definite Hermitian metric, not a metric degenerating to zero or infinity on C. □

Remark 5.3. The coefficient choice in (5.16) is particularly useful at the Lipschitz exponent $s = r^{1/2}$. At that exponent the product of the leading tangential and normal lower bounds is of constant order, so merely shrinking the neighborhood need not dominate a fixed mixed coefficient. Increasing M first supplies that domination. This is also why an argument that only says that the normal coefficient tends to infinity would not suffice: positivity requires the determinant inequality, not just the two diagonal inequalities.

5.4 The global auxiliary metric and its exact singularity

For global existence, suppose now that S is compact and projective, L is defined on S, and there are integers $m, a_0 > 0$ such that

$$A = L^{\otimes m} \otimes \mathcal{O}_S(-a_0 C) \quad \text{is ample.} \tag{5.18}$$

All line-bundle identities in what follows are holomorphic identities, not only numerical equivalences. Choose h_0 and k globally with the restrictions used above. Global extensions of the prescribed metrics can again be made by a smooth extension and a cut-off equal to one near C.

An ample bundle has a smooth strictly positive metric h_A. Explicitly, some $A^{\otimes N}$ is very ample and gives an embedding into projective space. Pull back the Fubini–Study metric on $\mathcal{O}(1)$ and take its Nth root. Its curvature is $\omega_A = N^{-1}\iota^*\omega_{\mathrm{FS}} > 0$.

The singular divisor metric $h_{[C]}$ on $\mathcal{O}_S(C)$ is characterized by $|s_C|^2_{h_{[C]}} = 1$ away from C. If $s_C = wv$, its local weight in the frame v is $\log|w|^2$, so its curvature current is $[C]$. Relative to k it satisfies

$$h_{[C]} = ke^{-\log r}.$$

Using the isomorphism $L^{\otimes m} = A \otimes \mathcal{O}_S(a_0 C)$, form the singular metric

$$h_*^m = h_A \otimes h_{[C]}^{a_0}.$$

This defines a metric on L: the positive mth root of the squared norm on $L^{\otimes m}$ is independent of the choice of frame. Its curvature current is

$$c_1(L, h_*) = \frac{1}{m}\omega_A + \frac{a_0}{m}[C]. \tag{5.19}$$

For the gluing we need a *global function* relative to h_0. Define a smooth real function F_0 on S by the ratio of two smooth metrics on the same line bundle:

$$h_A \otimes k^{a_0} = h_0^m e^{-F_0}.$$

The ratio is a globally defined positive smooth function, so its negative logarithm F_0 is globally well-defined. Then

$$h_* = h_0 e^{-\psi}, \qquad \psi = \frac{a_0}{m}\log r + \frac{F_0}{m}, \qquad \theta + dd^c\psi = \frac{1}{m}\omega_A + \frac{a_0}{m}[C]. \tag{5.20}$$

In particular, ψ is smooth on $S \setminus C$, tends to $-\infty$ uniformly on approach to C, and is strictly θ-plurisubharmonic off C. The coefficient of its logarithmic singularity is positive.

5.5 Gluing by a regularized maximum

Theorem 5.4 (Global Lipschitz existence)**.** *Let S be a smooth projective surface, let $C \subset S$ be a smooth connected curve with $C^2 < 0$, and let L be a holomorphic line bundle. Suppose $L|_C$ admits a smooth flat Hermitian metric and (5.18) holds for some positive integers m, a_0. Then L admits a positive definite Hermitian metric with locally Lipschitz weights and semipositive curvature current. The metric can be chosen smooth with strictly positive curvature on $S \setminus C$. Its curvature has bounded local potentials and minimal singularities in $c_1(L)$.*

Proof. Use Theorem 5.2 to choose U and M so that

$$u_{\mathrm{loc}} = Ms$$

has the asserted positivity on U. Choose a small number $b > 0$ such that $\{r \le 2b\} \subset U$. This is possible because S is compact, r is continuous with zero set exactly C, and the compact set $S \setminus U$ has a positive minimum of r whenever it is nonempty. Decrease b further if necessary. Construct ψ as in (5.20), and fix a regularization size $\varepsilon > 0$.

On the compact shell $\{b/2 \le r \le b\}$ the functions ψ and u_{loc} are smooth. Choose a real constant N such that

$$\psi + N > u_{\mathrm{loc}} + 2\varepsilon \quad \text{on } \{b/2 \le r \le b\}. \tag{5.21}$$

For instance, any $N > 2\varepsilon + \max_{b/2\le r\le b}(u_{\mathrm{loc}} - \psi)$ works; if the shell is empty the requirement is vacuous. The boundedness of F_0 and the positive coefficient of $\log r$ imply that $\psi + N$ tends uniformly to $-\infty$ as $r \downarrow 0$. Since $u_{\mathrm{loc}} \ge 0$, there exists a with $0 < a < b/4$ such that

$$\psi + N < u_{\mathrm{loc}} - 2\varepsilon \quad \text{on } \{0 < r \le 2a\}. \tag{5.22}$$

For complete uniformity, if $D_0 = \max_S F_0/m + N$, it suffices to choose $a < b/4$ so that $(a_0/m)\log(2a) + D_0 < -2\varepsilon$.

Let $\mathcal{M}_\varepsilon$ be the smooth regularized maximum from Lemma 2.3. Define a real function on S by

$$u(x) = \begin{cases} u_{\mathrm{loc}}(x), & r(x) \le a, \\ \mathcal{M}_\varepsilon(u_{\mathrm{loc}}(x), \psi(x) + N), & a < r(x) < b, \\ \psi(x) + N, & r(x) \ge b. \end{cases} \tag{5.23}$$

On a whole collar of the inner interface $r = a$, condition (5.22) makes the middle expression equal to u_{loc}. On a whole collar of the outer interface $r = b$, condition (5.21) makes it equal to $\psi + N$. Consequently the three pieces agree in neighborhoods of their interfaces. No regularity of the level hypersurfaces is needed for this pasting assertion.

We check positivity in local holomorphic frames of L. Let φ_0 be the weight of h_0. On the annulus, translation invariance of the regularized maximum gives

$$\varphi_0 + \mathcal{M}_\varepsilon(u_{\mathrm{loc}}, \psi + N) = \mathcal{M}_\varepsilon(\varphi_0 + u_{\mathrm{loc}}, \varphi_0 + \psi + N).$$

Both input functions are smooth and strictly plurisubharmonic there: the first by the local theorem and the second by (5.19) off C. By Lemma 2.3 and Remark 2.4, their regularized

maximum is smooth and strictly plurisubharmonic on the annulus. On the inner region positivity follows from the local theorem. On the outer region it follows from (5.19). The collars show these assertions fit together, proving that $\theta + dd^c u$ is positive globally and strictly positive away from C.

Near C, the function u equals Ms, hence is locally Lipschitz. On the complement it is smooth, including at both interfaces by the collar equalities. Thus u is a globally defined, locally Lipschitz, finite function on compact S, and is bounded. The metric $h = h_0 e^{-u}$ is positive definite with locally Lipschitz weights; it is smooth off C and has curvature $\theta + dd^c u$. Finally, Lemma 2.6 proves minimal singularities and bounded local potentials. This last step is a comparison with all θ-plurisubharmonic potentials, not a consequence of nefness alone. □

Remark 5.5. The gluing combines potentials relative to the same θ and uses the holomorphic identity (5.18). A partition of unity applied directly to plurisubharmonic weights would introduce uncontrolled derivative terms. Likewise, a merely numerical equality with an ample class would not by itself justify the metric identity used to define F_0. Both issues are avoided by the explicit construction above.

6 Applications to the original and modified examples

Corollary 6.1. *For the ruled surface and the line bundle L_0 of Assumption 3.1, there exists a positive definite Hermitian metric on L_0 with locally Lipschitz weights and semipositive curvature current. The metric is smooth and has strictly positive curvature on $X \setminus C$, and its curvature has bounded local potentials and minimal singularities. No neighborhood of C admits a C^1 semipositive Hermitian metric on L_0.*

Proof. The surface X is smooth and projective by Proposition 3.4. The curve C is smooth and connected, and $C^2 = -d < 0$, by Proposition 3.5. That proposition also provides the holomorphic trivialization $L_0|_C \cong \mathcal{O}_C$; its constant Hermitian metric is flat. Finally, Proposition 3.6 proves that the actual holomorphic line bundle

$$A = L_0^{\otimes 2} \otimes \mathcal{O}_X(-C)$$

is ample. These are all the hypotheses of Theorem 5.4, with $S = X$, $L = L_0$, $m = 2$, and $a_0 = 1$. That theorem proves the asserted existence and all its regularity and minimality properties. The class ξ is nonzero by assumption, so Theorem 4.3 gives nonexistence at the C^1 level, even on a neighborhood of C. □

Completion of the proof of Theorem 1.1. Part (i) is Propositions 3.4 and 3.5. The intersection, nefness, bigness, and null-locus assertions in part (ii) are Proposition 3.6, and the statement about all positive powers is Proposition 3.9. Parts (iii) and (iv) are precisely Corollary 6.1. □

6.1 The degree-one choice and the canonical-bundle choice

For an entirely specified degree-one instance, let B have genus two, choose $b_0 \in B$, and put $F = \mathcal{O}_B(b_0)$. Lemma 3.2 gives $H^1(B, F) \cong \mathbb{C}$, so choose any nonzero element. If $f_{b_0} = q^{-1}(b_0)$ denotes the actual fiber divisor, then

$$L_0 = \mathcal{O}_X(C + f_{b_0}), \qquad C^2 = -1, \qquad L_0^2 = 1. \tag{6.1}$$

The line-bundle equality in this display uses $F = \mathcal{O}_B(b_0)$. For a general degree-one line bundle F, replacing q^*F by a single fiber is only a numerical equivalence unless a corresponding holomorphic isomorphism has been chosen. Also, a general degree-one bundle on a genus-two curve need not have nonzero H^1. The choice just made avoids both qualifications.

For Koike's modified example, take again a genus-two curve and set $F = K_B$. Its degree is two. Serre duality on the curve gives

$$H^1(B, K_B) \cong H^0(B, \mathcal{O}_B)^* \cong \mathbb{C}$$

because B is compact and connected [Har77, Chapter III, §7]. A nonzero element therefore gives another nonsplit extension of the required form. In this case

$$L_0 = \mathcal{O}_X(C) \otimes q^* K_B, \qquad C^2 = -2, \qquad L_0^2 = 2. \tag{6.2}$$

Consequently Corollary 6.1 applies to both the original degree-one example and this degree-two example. Their numerical invariants differ; they are not being identified as the same ruled surface.

6.2 The bounded-potential question

The question in [Zha25, §4] concerns the existence of a positive current with bounded potentials and minimal singularities in the distinguished class for either example. We spell out exactly what the construction supplies. Fix a smooth metric h_0 on L_0 and set $\theta = c_1(L_0, h_0)$. The global function u constructed in (5.23) is bounded and satisfies

$$T = \theta + dd^c u \geq 0.$$

In a holomorphic frame, $T = dd^c(\varphi_0 + u)$, and $\varphi_0 + u$ is bounded on each relatively compact chart. For any other $v \in \mathrm{PSH}(X, \theta)$, compactness gives $\sup_X v < \infty$ unless $v \equiv -\infty$, in which case the comparison below is automatic. Since u is bounded below,

$$v \leq u + (\sup_X v - \inf_X u).$$

Thus T has minimal singularities by the defining comparison. These are exactly the two requested properties. Moreover its potentials are locally Lipschitz, a stronger continuity assertion. The construction provides an affirmative answer for the data of (6.1) and (6.2).

This conclusion is an implication of the proofs in this manuscript; it is not attributed to Grauert, Koike, or Zhang. In particular, Zhang's December 2025 version is cited as posing the question, not as already proving the bounded-potential statement. No assertion about publication priority of the present elementary construction is needed for this implication.

The assertion concerns existence of a representative with this regularity. Any two potentials with minimal singularities differ by a bounded function, by applying the defining comparison in both directions. Bounded difference alone does not imply that every minimal potential is continuous or Lipschitz; no such conclusion is made here.

6.3 Why the contracted surface is not Kähler

The analytic contraction from Proposition 3.11 is compatible with the existence of the Lipschitz current on X. In particular, one should not infer that the current supplies a Kähler form on the singular surface Y. The following argument establishes the opposite conclusion directly.

Proposition 6.2. *The normal compact surface Y obtained by contracting C is not Kähler in the usual sense for complex spaces: it has no Kähler form locally given by smooth strictly plurisubharmonic potentials in ambient holomorphic embeddings.*

Proof. Suppose such a Kähler form ω_Y exists. Pull it back along $\pi : X \to Y$. The result is a smooth real closed semipositive $(1,1)$-form β on X. To justify smoothness even on C, embed a neighborhood in Y into a domain in $\mathbb{C}^N$ and represent ω_Y there by $dd^c\Phi$, with Φ a smooth

strictly plurisubharmonic ambient function. Its pullback is $dd^c(\Phi \circ \pi)$, where the composition uses the holomorphic map into that ambient domain. This is a smooth form on X. These local pullbacks agree, and they are semipositive by the chain rule for the complex Hessian. They are strictly positive away from C, since π is biholomorphic there. Along TC, the form vanishes because $\pi|_C$ is constant. Therefore

$$\int_C \beta = 0, \qquad \int_f \beta > 0 \tag{6.3}$$

for every fiber f of q. For the second inequality, $f \setminus C$ is nonempty and β is strictly positive on its tangent line; its restriction to all of f is nonnegative.

We next identify its real cohomology class. The needed instance of the Leray–Hirsch theorem says that, for a $\mathbb{P}^1$-bundle and a class restricting to the generator of $H^2(\mathbb{P}^1, \mathbb{R})$, the cohomology of the total space is a free module over that of the base with basis that class and 1 [BT82, Theorem 5.11]. Here $c_1(L_0)$ restricts to that generator because $L_0|_f = \mathcal{O}_{\mathbb{P}^1}(1)$. Choose $\tau \in H^2(B, \mathbb{R})$ with $\int_B \tau = 1$. In degree two the theorem gives the direct-sum description

$$H^2(X, \mathbb{R}) = \mathbb{R}\, c_1(L_0) \ \oplus \ \mathbb{R}\, q^*\tau.$$

Hence $[\beta] = ac_1(L_0) + bq^*\tau$ for real numbers a, b. Pairing with C gives $b = 0$, since $L_0 \cdot C = 0$ and $\int_C q^*\tau = 1$. Pairing with f then gives $a = \int_f \beta > 0$. Thus $[\beta/a] = c_1(L_0)$.

The smooth projective surface X is compact Kähler. The $\partial\bar\partial$-lemma on a compact Kähler manifold says that a real closed $(1,1)$-form with zero real de Rham class is dd^c of a smooth real function [Dem12b, Chapter VI, Lemma 8.6]. Apply it to $\beta/a - \theta$, where θ is the curvature of any smooth Hermitian metric h_0 on L_0. There is a smooth real function v such that

$$\beta/a - \theta = dd^c v.$$

The metric $h_0 e^{-v}$ on L_0 is smooth and has semipositive curvature β/a. This contradicts Theorem 4.3. Hence ω_Y cannot exist. □

Remark 6.3. The metric constructed here has near C a weight of the form $\varphi_0 + M|w|e^{\lambda/2}$. Its failure to be C^1 is visible on a real normal line $w = t$: the two one-sided derivatives of its cusp term at $t = 0$ are $Me^{\lambda(z,0)/2}$ and $-Me^{\lambda(z,0)/2}$. Adding the smooth function φ_0 does not remove this jump. The obstruction theorem says much more than this observation about our formula: no different semipositive metric can have C^1 weights near C. Conversely, the non-Kählerness of Y does not contradict the positive current on X: the latter is not a smooth Kähler form near C, and the construction gives no smooth strictly plurisubharmonic potentials descending to the contracted point.

7 Historical context and the scope of the conclusion

The geometric construction, the obstruction to smooth semipositivity, and the bounded-potential question arose at different times. We distinguish these statements, and also distinguish Grauert's degree-one example from Koike's degree-two modification. In the notation used here,

$$0 \longrightarrow F \longrightarrow E \longrightarrow \mathcal{O}_B \longrightarrow 0, \qquad X = \mathbb{P}_{\text{lines}}(E), \qquad C = \mathbb{P}_{\text{lines}}(F), \qquad L_0 = \mathcal{O}_X(C) \otimes q^*F,$$

where $q : X \to B$ is the projection, $\deg F = d > 0$, and the extension class $\xi \in H^1(B, F)$ is nonzero. These conventions identify the normal bundle as $N_{C/X} = F^{-1}$ and give $C^2 = -d$.

7.1 Grauert's construction and the degree-one choice

Grauert's example occurs in [Gra62, §4, paragraph 8(d), pp. 365–366]. Its starting data are a genus-two curve B, a degree-one line bundle F, and a nontrivial affine-line bundle modeled on F. The affine bundle is an F-torsor: differences of local sections are sections of F, but its transition functions also contain translations. Compactifying each affine fiber by one point gives the ruled surface above; the added section is C. Thus the affine bundle itself is not a locally free sheaf, whereas E is a rank-two locally free sheaf.

The cohomological hypothesis $H^1(B, F) \neq 0$ is essential. It does not follow merely from $\deg F = 1$. One explicit valid choice is $F = \mathcal{O}_B(b)$ for a point $b \in B$: here $h^0(B, F) = 1$, and Riemann–Roch gives $h^1(B, F) = 1$. For comparison, a degree-one bundle outside the effective locus has both groups zero. The qualification concerning $H^1(B, F)$ is also recorded in note T28 of the supplied English translation of Grauert's paper. Choosing $\mathcal{O}_B(b)$ resolves it without altering the intended example.

Since $C^2 = -1$, Grauert's contraction theorem produces a proper holomorphic map from X to a normal compact complex surface, contracting precisely C to one point. The relevant contraction results are proved in [Gra62, §3 and §4, paragraph 8(e)]; a modern surface treatment is [BHPV04, Chapter III, §2]. The contracted point is singular, despite $C^2 = -1$, because C has genus two. Grauert proves that the target is not projective: a hyperplane section avoiding the contracted point would lift to a compact multisection of the affine bundle, and the fiberwise average of its points would give a section of the nontrivial torsor. In particular, Grauert's original argument concerns analytic contraction and nonprojectivity; it does not establish an obstruction to continuous semipositive metrics.

7.2 The questions of Filip–Tosatti and the obstruction of Koike

Filip–Tosatti revisit this surface in [FT18, §2, especially Problem 2.2, p. 2989]. For an ample line bundle A they consider, in additive divisor notation,

$$A - \frac{A \cdot C}{C^2}\, C.$$

Its intersection with C is zero, while the class is nef and big. Their question asks for the failure of *Hermitian semipositivity*, meaning the existence of a smooth Hermitian metric with semipositive curvature. When $C^2 = -1$ this displayed expression is integral. If the construction is written with $C^2 = -d$ for general d, it is initially a rational divisor class; denominators must be cleared before it is treated as a line bundle. This issue does not arise for the explicitly defined integral bundle L_0. Filip–Tosatti also explain that the contracted surface is non-Kähler, referring to [Moi75]. That assertion is not needed for the metric construction in the present manuscript.

Koike develops first-order obstructions to linearizing a line bundle along a compact submanifold in [Koi21]. His Theorem 1.4 shows that a nonzero class

$$u_1(C, X, L) \in H^1(C, N_{C/X}^{-1})$$

precludes smooth semipositivity when C is compact Kähler and $L|_C$ is topologically trivial. This extends an obstruction-theoretic viewpoint originating in Ueda's study of curves with topologically trivial normal bundle [Ued83]. The normal bundle of the present curve is negative, so Ueda's original hypotheses are not themselves the hypotheses of the application here.

In [Koi21, Example 4.1, pp. 2262–2263], Koike takes a genus-two curve and $F = K_B$. Thus his explicit model has $d = 2$, $C^2 = -2$, and $L_0^2 = 2$. A cocycle obtained from the hyperbolic metric represents a nonzero class in $H^1(B, K_B)$, and the first-order obstruction

of L_0 is nonzero. This proves that this modified Grauert example is nef and big but has no smooth semipositive metric. It is important not to identify its degree-two data with Grauert's original degree-one data.

7.3 Zhang's formulation and the conclusion proved here

In [Zha25], Zhang returns to the degree-one construction and to the precise line bundle $L_0 = q^*F \otimes \mathcal{O}_X(C)$. The calculation in §3 identifies its first obstruction with the nontrivial affine-bundle class and applies Koike's theorem to exclude a smooth semipositive metric. Section 4 asks whether the first Chern class on the original or modified Grauert example admits a positive current with minimal singularities and bounded local potentials. The version cited here is arXiv:2512.23446v1, submitted on 29 December 2025; this formulation is attributed to that specific version.

Theorem 1.1 and the global construction in Theorem 5.4 give an affirmative answer for the data treated here: L_0 has a positive-curvature Hermitian metric with locally Lipschitz weights. This includes both the original degree-one case and the degree-two canonical-bundle case, and more generally every positive-degree F and nonzero extension class satisfying the stated hypotheses. Theorem 4.3 also excludes a metric with C^1 weights and semipositive curvature. These regularity conclusions are deductions proved in this manuscript; they are not statements being attributed to Grauert, Filip–Tosatti, Koike, or Zhang. No claim of bibliographic priority is intended.

Boundedness implies minimal singularities in this situation for an elementary reason. Fix a smooth reference metric and write a positive-curvature metric as a global quasi-plurisubharmonic weight φ. Every competing such weight ψ is bounded above on the compact surface. If φ is bounded below, then $\psi \leq \varphi + O(1)$, which is the defining comparison for minimal singularities. This terminology and its general framework are developed in [DPS01, §1] and [BEGZ10]. Thus the bounded-potential conclusion concerns the same metric and class, without a separate existence theorem for minimal singularities.

7.4 What the standard tools do and do not supply

The projectivity of X follows by algebraizing the extension on the projective curve using GAGA [Ser56]. The projective-bundle and ruled-surface calculations are standard [Har77, Chapters II and V]; here the convention of parametrizing lines is fixed explicitly. Ampleness of the auxiliary bundle used in the construction is checked by the surface numerical criterion [Nak63, Laz04]. These algebraic facts provide a smooth positive metric away from the exceptional-curve difficulty; they do not give smooth semipositivity of L_0.

The analytic patching uses maxima of plurisubharmonic functions and removable singularities for locally bounded-above plurisubharmonic functions. Suitable precise versions are [Dem12b, Chapter I, Lemma 5.17 and Theorem 5.24]; regularized maxima and their preservation of Levi lower bounds appear in [Dem12b, Chapter I, Lemma 5.18], within the regularization theory of [Ric68]. The decisive local estimate and the C^1 obstruction are supplied in the present proofs. In particular, these regularization results do not authorize smoothing the final metric across C while preserving its semipositive curvature.

There is no general implication from nefness to bounded semipositive weights. The nonsplit extension over an elliptic curve in [DPS94, Example 1.7] gives a nef, non-big tautological bundle; [Koi15, Corollary 4.1 and Example 4.2] explains its unbounded minimal-metric behavior and a related nef and big threefold example. Koike's positive results for negative curves in that paper impose, among other conditions, a holomorphic tubular-neighborhood hypothesis. Such a hypothesis is not assumed here and cannot be silently substituted for the nontrivial affine transitions.

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